Relational Dualities and Bisimulation

Piotr Kozicki and G. A. Kavvos

University of Bristol

Abstract. The Kripke semantics of various logics arise through categorical dualities between a category of frames and frame maps, and a category of algebras and logical homomorphisms. We argue that these dualities can be extended to a relational level, associating well-behaved relations between frames (e.g. bisimulations) to relations between algebras of predicates. This is substantiated by constructing relational versions of the Tarski duality (for classical propositional logic) and Thomason duality (for classical modal logic).

1 Introduction

It is a well-known secret that relational semantics (e.g. Kripke semantics) arise from *dualities*, i.e. categorical equivalences of the form $\mathcal{C}^{\mathsf{op}} \simeq \mathcal{D}$. The idea is that $\mathcal{C}$ is a 'spatial' category whose objects are *frames*, and $\mathcal{D}$ is an 'algebraic' category whose objects are *algebras*, which support logical structure.

For example, truth table semantics for classical propositional logic arise from the *Tarski duality* $\mathbf{Set}^{\mathsf{op}} \simeq \mathbf{CABA}$ between the category of sets and the category of *complete atomic Boolean algebras* (CABAs) [23, §1]. The Kripke semantics of classical modal logic [24, 26] arise from the *Thomason duality* $\mathbf{Frm}^{\mathsf{op}} \simeq \mathbf{CABAO}$ between the category of Kripke frames and the category of CABAs with operators (i.e. modalities) [37] [23, §2.4]. The Kripke semantics of intuitionistic logic [25] arises from the duality $\mathbf{Pos}^{\mathsf{op}} \simeq \mathbf{PrAlgLatt}$ between posets and prime algebraic lattices [30, 21]. Famous dualities of this form include Stone duality (for Boolean algebras) [19], Jónsson-Tarski duality [20, 12, 32] (for Boolean algebras with operators), and Esakia duality [10] (for Heyting algebras). See the chapter by Kishida [23] for further references.

The overwhelming majority of work in this area has not paid attention to the *morphisms* of these categories.[1] In many cases the morphisms $f : W \to W'$ of $\mathcal{C}$ can be thought of as 'truth-preserving': they map worlds to worlds in a way that preserves (and often reflects) truth, e.g. $w \vDash \varphi$ implies $f(w) \vDash \varphi$. The corresponding morphism $f^* : \mathbb{C}(W') \to \mathbb{C}(W')$ of $\mathcal{D}$ then preserves the logical structure, e.g. $f^*(\varphi \wedge \psi) = f^*(\varphi) \wedge f^*(\psi)$ in the case of conjunction.

Obtaining this property in the case of intuitionistic and modal logic requires strong properties on morphisms. For example, in the Kripke semantics of modal logic a map $f : W \to W'$ of posets preserves truth if and only if the corresponding map $f^* : \mathcal{P}(W') \to \mathcal{P}(W)$ preserves the modality $\Box$ [23, §2.4]. It is possible

[1] One exception is the work of Abramsky [3] on domain theory in logical form.

to characterise such maps as the *open maps*,[2] i.e. functions whose graph is a *bisimulation* between Kripke frames.

This observation raises an interesting question: what happens if we relax the morphisms of $\mathcal{C}$, requiring them to merely be *relations* instead of functions? Spurred by the example of intuitionistic logic we expect the morphisms of $\mathcal{C}$ to have some particular relational property (e.g. to be bisimulations). Likewise, we expect the morphisms of $\mathcal{D}$ to be relations between algebras of predicates, but what properties they should have is somewhat unclear.

This paper attempts to shed some light on this question by developing basic examples of such *relational dualities*. This will be done chiefly from a computational logic point of view: a bisimulation $R \subseteq W \times W'$ can be thought of as a correspondence between the states of two systems, with $w \, R \, w'$ meaning that w and w' have the same behaviours. A relational duality will then induce a correspondence $\mathbb{C}(R) \subseteq \mathbb{C}(W) \times \mathbb{C}(W')$ between *predicates* on these systems, with $\varphi \, \mathbb{C}(R) \, \psi$ meaning that systems that satisfy φ correspond under R to systems that satisfy ψ. The ultimate goal is to develop a formal system for synthetically reasoning about *relations between formulas over different systems*. It is not difficult to imagine that this might have applications in program logics, with bisimulation inducing relations between formulae over different primitives.

Roadmap The existence of relational dualities is not to be doubted: perhaps the simplest one is the obvious self-duality $\mathbf{Rel}^{\mathsf{op}} \simeq \mathbf{Rel}$. Instead, this paper aims to to justify the claim that that there are relational dualities with non-trivial logical content. This will be achieved by *extending* the two simplest 'discrete' dualities, namely the Tarski duality $\mathbf{Set}^{\mathsf{op}} \simeq \mathbf{CABA}$, which gives rise to classical propositional logic,[3] and the Thomason duality $\mathbf{Frm}^{\mathsf{op}} \simeq \mathbf{CABAO}$, which gives rise to the classical modal logic $\mathbf{K}$.

After recalling some preliminary definitions to fix notation (§2), we briefly recap the elements of Tarski duality (§3). Then, we show how it can be extended to a *relational Tarski duality* (§4); this will be achieved by using one of the simplest ways of lifting a relation between sets to a relation on powersets, namely the *lower relation* (∀∃). We precisely characterise the relations between CABAs that we obtain this way, and establish an equivalence. Furthermore, we show that this relational Tarski duality is an extension of the usual Tarski duality.

Following this, we will show how the same lifting gives rise to an extensional version of the Thomason duality (§5). This is somewhat more challenging: while Tarski duality acts on arbitary functions, the Thomason duality uniquely associates open maps (= functional bisimulations) with modal complete Boolean homomorphisms. We show that this has an analogue on the relational level, putting (bi)simulations in perfect correspondence with what we call *(bi)simulatory relations* between CABAs. Finally We show that this *relational Thomason duality* extends the ones originally given by Thomason.

[2] Open maps are also called *p-morphisms* [6, §2.3] or *bounded morphisms* [5, §2.1].

[3] Tarski duality actually induces an *infinitary* classical propositional logic, as the powerset $\mathcal{P}(X)$ supports arbitrary joins and meets. However we do not envision writing any infinitary formulae.

2 Preliminaries

A *relation* $R : A \twoheadrightarrow B$ from a set A to a set B is a subset $R \subseteq A \times B$. We write $a\,R\,b$ to mean $(a, b) \in R$. If $R : A \twoheadrightarrow B$ and $S : B \twoheadrightarrow C$ their composition $R\,;S : A \twoheadrightarrow C$ is defined in the usual manner, i.e. $a\,(R\,;S)\,c$ if there exists some $b \in B$ with $a\,R\,b\,S\,c$. Note that we write the composition of relations using diagrammatic order. Given a relation $R : A \twoheadrightarrow B$ we define its *opposite* $R^\dagger : B \twoheadrightarrow A$ by $b\,R^\dagger\,a$ iff $a\,R\,b$. Sets and relations form a category **Rel** under composition, with the identity $\mathsf{Id}_A : A \twoheadrightarrow A$ given by $a_1\,\mathsf{Id}_A\,a_2$ iff $a_1 = a_2$. Moreover, the opposite construction extends to a functor $(-)^\dagger : \mathbf{Rel}^{\mathsf{op}} \to \mathbf{Rel}$ which induces a formal duality $\mathbf{Rel}^{\mathsf{op}} \simeq \mathbf{Rel}$ given by taking the opposite of a relation. This shows that **Rel** is a *dagger category* [14].

A partial order $(P, \sqsubseteq_P)$ is a set P equipped with a relation $\sqsubseteq_P : P \twoheadrightarrow P$ which is reflexive, transitive, and anti-symmetric. We will commonly refer to partial orders by their carriers (P, Q, ...) and often elide the subscript of $\sqsubseteq$. A function $f : P \to Q$ is *monotonic* just if $x \sqsubseteq_P y$ implies $f(x) \sqsubseteq_Q f(y)$.

A *lattice* is a partial order that has all finite joins and meets. We will use the letters $\mathcal{L}, \mathcal{L}', \ldots$ to denote lattices. A *complete lattice* $\mathcal{L}$ is a partial order that has all joins, which also implies that it has all meets.

A pair of monotonic functions $f : P \to Q$ and $g : Q \to P$ is an *adjoint pair* (or *adjunction*) just if $f(x) \sqsubseteq_Q y$ iff $x \sqsubseteq_P g(y)$. This is often denoted as $f \dashv g$. It is a consequence that f preserves all joins and g preserves all meets. The *adjoint functor theorem* [8, §7.34] [19, §I.4.2] says that if $f : P \to Q$ is monotonic, P is a complete lattice, and f preserves all joins, then f has a right adjoint $g : Q \to P$. By duality, if $g : Q \to P$ is monotonic, Q is a complete lattice, and g preserves all meets, then g has a left adjoint $f : P \to Q$.

We refer the reader to [8] for further background in orders and lattices.

3 A Primer on Tarski Duality

We recap the Tarski duality $\mathbf{Set}^{\mathsf{op}} \simeq \mathbf{CABA}$, and show how it induces an (infinitary) classical propositional logic.

Recall that a Boolean algebra is a distributive lattice $\mathcal{B}$ in which every element $x \in \mathcal{B}$ has a *complement* $\neg x \in \mathcal{B}$ for which $x \vee \neg x = \top$ and $x \wedge \neg x = \bot$.

For any set X, its powerset $\mathcal{P}(X)$ is a Boolean algebra [8, §5.2]. In fact, it is a *complete* Boolean algebra, i.e. a complete lattice, with joins and meets of $(S_i)_{i\in I}$ given by $\bigcup_{i\in I} S_i$ and $\bigcap_{i\in I} S_i$ respectively. Moreover, $\mathcal{P}(X)$ is *atomic*, in the following sense.

Definition 1 (Atom). *Let $(P, \sqsubseteq)$ be a partial order with a bottom element $\bot$. $a \in \mathcal{L}$ is an* atom *of $\mathcal{L}$ if $a \neq \bot$ and $x \sqsubseteq a$ implies that either $x = \bot$ or $x = a$. We write $\mathsf{At}(P)$ for the set of atoms of P.*

Definition 2 (Atomic complete lattice). *A complete lattice $\mathcal{L}$ is* atomic *just if every element is equal to the join of atoms below it. In symbols, for every $x \in \mathcal{L}$ we have $x = \bigsqcup \{a \in \mathsf{At}(\mathcal{L}) \mid a \sqsubseteq x\}$.*

Definition 3. *A* complete atomic Boolean algebra (CABA) *is a complete Boolean algebra which is moreover atomic.*

For any set X, the powerset $\mathcal{P}(X)$ is a CABA: its atoms are exactly the singleton sets $\{x\}$ for $x \in X$; every $S \subseteq X$ can be reconstructed as $S = \bigcup_{x \in S} \{x\}$.

Let $\mathcal{B}$ and $\mathcal{B}'$ be CABAs. A *morphism of CABAs* $f^* : \mathcal{B} \to \mathcal{B}'$ is a monotonic function that preserves all meets and joins. This definition has deceptively many implications. First, as $\mathcal{B}$ and $\mathcal{B}'$ are Boolean algebras, every such $f^* : \mathcal{B} \to \mathcal{B}'$ is a *complete Boolean homomorphism*, i.e. preserves *all* the Boolean operations. Second, by the adjoint functor theorem the map $f^* : \mathcal{L} \to \mathcal{L}'$ has both a left and a right adjoint:

$$\begin{tikzcd} \mathcal{B}' \arrow[rr, bend left, "f_!"] \arrow[rr, bend right, "f_*"'] & \arrow[phantom, "\perp"] & \mathcal{B} \arrow[ll, "f^*" description] \end{tikzcd}$$

CABAs and their morphisms form a category **CABA**.

Given a function $f : X \to Y$ we can define a function $f^* : \mathcal{P}(Y) \to \mathcal{P}(X)$ by $f^*(B) = \{x \in X \mid f(x) \in B\}$ [28, 29, 23]. This function preserves all joins and meets, and hence it is a morphism of CABAs. By the adjoint functor theorem, f^* has both a left and a right adjoint, given by

$$\begin{aligned} f_!(A) &= \{y \in Y \mid \exists x \in X.\ f(x) = y \wedge x \in A\} &&= \{f(x) \mid x \in A\} \\ f_*(A) &= \{y \in Y \mid \forall x \in X.\ f(x) = y \Rightarrow x \in A\} &&= \{y \in Y \mid f^*(\{x\}) \subseteq A\} \end{aligned}$$

Thus, we obtain a functor $\mathcal{P} : \mathbf{Set}^{\mathsf{op}} \longrightarrow \mathbf{CABA}$.

Lemma 1. *Let $f^* : \mathcal{B} \to \mathcal{B}'$ be a morphism of CABAs. Then its left adjoint $f_! : \mathcal{B}' \to \mathcal{B}$ maps atoms to atoms.*

We can thus define a functor $\mathsf{At} : \mathbf{CABA} \to \mathbf{Set}^{\mathsf{op}}$, which maps a CABA $\mathcal{B}$ to its set of atoms $\mathsf{At}(\mathcal{B})$, and use Lemma 1 to map a morphism $f^* : \mathcal{B} \to \mathcal{B}'$ to the restriction $f_!|_{\mathsf{At}(\mathcal{B})} : \mathsf{At}(\mathcal{B}) \to \mathsf{At}(\mathcal{B}')$ of its left adjoint to atoms. This is a pseudo-inverse to $\mathcal{P}$, and we obtain a categorical equivalence.

Theorem 1 (Tarski duality). $\mathbf{Set}^{\mathsf{op}} \simeq \mathbf{CABA}$

The gist is that every CABA $\mathcal{B}$ is isomorphic to the powerset $\mathcal{P}(\mathsf{At}(B))$ of its atoms: its elements are uniquely determined by the atoms below them.

This duality induces a simple classical logic. Let W be a set of *worlds*. This set corresponds to the powerset $\mathcal{P}(W)$, which can be seen as a set of *predicates*. A world $w \in W$ satisfies the predicate $\varphi \in \mathcal{P}(W)$ just if $w \in \varphi$. As $\mathcal{P}(W)$ is a Boolean algebra, predicates are closed under all Boolean operations.[4] Individual worlds $w \in W$ correspond to singleton predicates $\{w\} \in \mathcal{P}(W)$ which uniquely

[4] The fact $\mathcal{P}(W)$ is a *complete* Boolean algebra means that predicates are also closed under *infinite* conjunction $\bigwedge_{i \in I} \varphi_i$ and disjunction $\bigvee_{i \in I} \varphi_i$.

characterise them. If we interpret W as as the *set of states* of a computer, a logic like this has been in continuous use since the pioneering work of Dijkstra [9].

Tarki duality has consequences for logic. For example, it is closely related to *completeness* of classical propositional logic. Let Var be a *finite* set of propositional variables, and let $\mathbb{BA}(\mathsf{Var})$ be the free Boolean algebra on these variables (i.e. propositional formulae quotiented under logical equivalence). As this Boolean algebra is finite, it is complete and atomic [8, §5.5], and hence a CABA. It is thus isomorphic to the powerset $\mathcal{P}(\mathsf{At}(\mathbb{BA}(\mathsf{Var})))$. It can be shown that the atoms of $\mathbb{BA}(\mathsf{Var})$ are the so-called *minterms*, i.e. conjunctions of literals that mention every variable in Var, which correspond uniquely to Boolean valuations. Thus, two formulas are equivalent iff they have the same set of valuations. This shows that the traditional *truth table semantics* of Wittgenstein [38] is just a Kripke semantics for classical propositional logic [23, §1]. However, most textbook presentations do not explicitly display the set of worlds W, as any two worlds that satisfy exactly the same propositions are indistinguishable.

Certain strands of work on logic and duality [2, 1, 3, 21, 22] have proposed taking the morphisms seriously. In Tarski duality, a function $f : W \to W'$ between worlds induces a complete Boolean homomorphism $f^* : \mathcal{P}(W') \to \mathcal{P}(W)$. These preserve all logical structure—e.g. $f^*(\varphi \wedge \psi) = f^*(\varphi) \wedge f^*(\psi)$. In the free case f^* can be pushed through all connectives until it reaches a propositional variable p, where it acts as $f^{-1}(p)$. Hence, a function $f : W \to W'$ allows us to map predicates on W' to predicates on W in a logic-preserving manner.

4 A Relational Tarski Duality

This last point raises a question: if we were to replace a function $f : W \to W'$ with an arbitrary *relation* $R : W \to\!\!\!\!\!\to W'$ between sets of worlds, what would the corresponding connection between the sets of predicates $\mathcal{P}(W)$ and $\mathcal{P}(W')$ be?

It is possible to obtain such a duality by replacing **CABA** with larger category $\mathbf{CABA}_{\bigvee}$, whose morphisms only preserve joins, and then show that $\mathbf{Rel} \simeq \mathbf{CABA}_{\bigvee}$ [20, 34] [23, §2.3]. Indeed, if $h : \mathcal{P}(X) \to \mathcal{P}(Y)$ preserves all joins, then writing any set as a union of singletons, we see that f is completely determined by a function $X \to \mathcal{P}(Y)$, i.e. a relation $X \to\!\!\!\!\!\to Y$. Hofmann and Nora [16, §4.5] show that this duality is a special case of a general construction. However, these more general morphisms do *not* preserve the logical structure of predicates; for example it may be that $h(\varphi \wedge \psi) \neq h(\varphi) \wedge h(\psi)$. Instead, we would like to find a duality whose logical side consists of a category of algebras of predicates and *relations between these predicates*.

To achieve that, we consider a simple way of *lifting* relations to powersets.

Definition 4 (Lower Relation). *Given a relation $R : X \to\!\!\!\!\!\to Y$, the associated* lower relation $\mathfrak{L}(R) : \mathcal{P}(X) \to\!\!\!\!\!\to \mathcal{P}(Y)$ *is defined by*

$$S \,\mathfrak{L}(R)\, T \;\stackrel{\text{def}}{\equiv}\; \forall s \in S.\ \exists t \in T.\ s \,R\, t.$$

This manner of lifting relations has a long history. It is one-half of the *Egli-Milner lifting*, which first appeared in the powerdomain literature [31, 13]. It then resurfaced in the coalgebra literature [36], where post-fixed-points amount to *simulations* between transition systems [18, §3].

$\mathfrak{L}$ maps a relation between sets to a relation between CABAs. However, it is *not* a functor $\mathbf{Rel} \longrightarrow \mathbf{Rel}$: it maps the identity relation $\mathsf{id} : X \nrightarrow X$ to the containment relation $\subseteq \,:\, \mathcal{P}(X) \nrightarrow \mathcal{P}(X)$. We will therefore define a category **CABARel** so that $\mathfrak{L}$ becomes a functor $\mathbf{Rel} \longrightarrow \mathbf{CABARel}$.

Definition 5 (Directionally atomic relation).

1. *A relation $R : P \nrightarrow Q$ between two partial orders is a* bimodule *just when $p' \sqsubseteq p \mathrel{R} q \sqsubseteq q'$ implies $p' \mathrel{R} q'$.*
2. *A relation $R : \mathcal{L} \nrightarrow Y$ where $\mathcal{L}$ is a complete lattice is* left-disjunctive *just if $a_i \mathrel{R} b$ for all $i \in I$ implies $\left(\bigsqcup_{i\in I} a_i\right) \mathrel{R} b$.*
3. *A relation $R : \mathcal{B} \nrightarrow \mathcal{B}'$ between CABAs is* atomic-founded *when for any atom $a \in \mathcal{B}$, if $a \mathrel{R} b$ then there exists an atom $b' \sqsubseteq b$ with $a \mathrel{R} b'$.*
4. *A relation $R : \mathcal{B} \nrightarrow \mathcal{B}'$ between two CABAs is* directionally atomic *just if it is left-disjunctive, atomic-founded, and a bimodule.*

Proposition 1. *If $R : \mathcal{B} \nrightarrow \mathcal{B}'$ and $S : \mathcal{B}' \nrightarrow \mathcal{B}''$ are directionally atomic, then so is their composition $R \,;\, S : \mathcal{B} \nrightarrow \mathcal{B}''$.*

Proof. First, we show that $R \,;\, S : X \nrightarrow Z$ is left-disjunctive. Suppose that $x_i \,(R \,;\, S)\, z$ for all $i \in I$. Then for each $i \in I$ there exists some $y_i \in Y$ such that $x_i \mathrel{R} y_i$ and $y_i \mathrel{S} z$. Define $u \stackrel{\text{def}}{=} \bigsqcup_{i\in I} y_i$. Then, as R is a bimodule, $x_i \mathrel{R} u$ for all $i \in I$, and as R is left-disjunctive, $(\bigsqcup_{i\in I} x_i) \mathrel{R} u$. As S is left-disjunctive, $u \mathrel{S} z$. Hence $(\bigsqcup_{i\in I} x_i) \mathrel{R \,;\, S} z$.

Second, $R \,;\, S : X \nrightarrow Z$ is atomic-founded. Suppose that $x \in X$ is an atom, and that for $z \in Z$ we have $x \,(R \,;\, S)\, z$. Then there exists some $y \in Y$ for which $x \mathrel{R} y$ and $y \mathrel{S} z$. As R is atomic-founded we know that there exists some atom $y' \sqsubseteq y$ for which $x \mathrel{R} y'$ and because S is a bimodule we have that $y' \mathrel{S} z$. But S is also atomic-founded, so there exists some atom $z' \sqsubseteq z$ for which $y' \mathrel{S} z'$. Therefore $x \,(R \,;\, S)\, z'$.

Third, we show that $R \,;\, S : X \nrightarrow Z$ is a bimodule. Suppose that $x' \sqsubseteq x$ and $z \sqsubseteq z'$, and that $x \,(R \,;\, S)\, z$, that is, there exists some $y \in Y$ for which $x \mathrel{R} y$ and $y \mathrel{S} z$. Because R is a bimodule we know that $x' \mathrel{R} y$ and because S is a bimodule we know that $y \mathrel{S} z'$, therefore $x' \,(R \,;\, S)\, z'$.

Proposition 2. *$\sqsubseteq \,:\, \mathcal{B} \nrightarrow \mathcal{B}$ is directionally atomic, and is an identity for composition of directionally atomic relations.*

Proof. First, we show that $\sqsubseteq$ is left-disjunctive. Suppose that $x_i \sqsubseteq y$ for any $i \in I$. Then $\bigsqcup_{i\in I} x_i \sqsubseteq y$, simply from the definition of a supremum. Second, $\sqsubseteq$ is atomic-founded. Let $x \in \mathcal{L}$ be an atom and suppose that $x \sqsubseteq y$. Then $x \sqsubseteq x$ by reflexivity, and $x \sqsubseteq y$ by assumption. Third, $\sqsubseteq$ is a bimodule, simply by transitivity. Finally, let $R : X \nrightarrow Y$ be directionally atomic. Then R is a bimodule, i.e. $R; \sqsubseteq \,=\, \sqsubseteq ;R = R$.

We can therefore define a category **CABARel** with CABAs as objects, and directionally atomic relations as morphisms.

Lemma 2. *For a relation $R : X \nrightarrow Y$ between sets, $\mathfrak{L}(R) : \mathcal{P}(X) \nrightarrow \mathcal{P}(Y)$ is directionally atomic.*

Proof. First, we show that the lower relation is left-disjunctive. Suppose for every $i \in I$ we have $A_i \,\mathfrak{L}(R)\, B$. Then for any $a \in \bigcup_{i\in I} A_i$ there must exist some $i \in I$ for which $a \in A_i$, and hence a $b \in B$ with $a\,R\,b$. Hence $\left(\bigcup_{i\in I} A_i\right) \mathfrak{L}(R)\, B$.

Second, we show that the lower relation is atomic-founded. Let $\{a\} \subseteq X$ be an atom in $\mathcal{P}(X)$. Then if $\{a\}\,\mathfrak{L}(R)\,B$ there exists some $b \in B$ for which $a\,R\,b$. Hence $\{a\}\,\mathfrak{L}(R)\,\{b\}$.

Finally, it is evident that $\mathfrak{L}(R)$ is a bimodule.

Lemma 3. $\mathfrak{L}$ *is a functor* **Rel** $\longrightarrow$ **CABARel**.

Proof. By Lemma 2 every $\mathfrak{L}(R)$ is a morphism in **CABARel**.

First, we show that $\mathfrak{L}$ preserves the identity. If $\mathsf{id} : X \nrightarrow X$ then $\mathfrak{L}(id)$ is just the relation $\subseteq$, which is the identity $\mathcal{P}(X) \nrightarrow \mathcal{P}(X)$ in **CABARel**.

Second, we show that $\mathfrak{L}$ preserves composition. Let $R : X \nrightarrow Y$ and $S : Y \nrightarrow Z$ be relations. Suppose $A\,\mathfrak{L}(R\,;S)\,C$. For any $a \in A$ define

$$B_a \stackrel{\text{def}}{=} \{b_a \in Y \mid a\,R\,b_a \wedge \exists c \in C.\ b_a\,S\,c\}\,, \qquad\qquad B \stackrel{\text{def}}{=} \bigcup_{a\in A} B_a$$

Then $A\,\mathfrak{L}(R)\,B$ and $B\,\mathfrak{L}(S)\,C$, so $A\,\mathfrak{L}(R)\,;\mathfrak{L}(S)\,C$.

For the converse, suppose instead that $A\,\mathfrak{L}(R)\,;\mathfrak{L}(S)\,C$, so there exists some B with $A\,\mathfrak{L}(R)\,B\,\mathfrak{L}(S)\,C$. Then for every $a \in A$ there must exist $b \in B$ and $c \in C$ with $a\,R\,b\,S\,c$. Hence $A\,\mathfrak{L}(R\,;S)\,C$.

Lemma 4. *The functor* $\mathfrak{L}$: **Rel** $\longrightarrow$ **CABARel** *is faithful.*

Proof. Suppose $R, S : X \nrightarrow Y$ are two relations and that $\mathfrak{L}(R) = \mathfrak{L}(S)$. Then $x\,R\,y$ iff $\{x\}\,\mathfrak{L}(R)\,\{y\}$ iff $\{x\}\,\mathfrak{L}(S)\,\{y\}$ iff $x\,S\,y$. Hence $R = S$.

Lemma 5. *The functor* $\mathfrak{L}$: **Rel** $\longrightarrow$ **CABARel** *is full. In other words, any directionally atomic $R : \mathcal{P}(X) \nrightarrow \mathcal{P}(Y)$ is $\mathfrak{L}(S)$ for some relation $S : X \nrightarrow Y$.*

Proof. We will prove that $R = \mathfrak{L}(R_S)$ where $S_Q : X \nrightarrow Y$ is $x\,S_Q\,y \stackrel{\text{def}}{\equiv} \{x\}\,R\{y\}$.

Suppose that $A\,R\,B$. Then for any $a \in A$ we must have that $\{a\}\,R\,B$, as R is a bimodule. But then we must also have that there exists some $b \in B$ such that $\{a\}\,R\,\{b\}$, because R is atomic-founded, i.e. $a\,S\,b$. Therefore $A\,\mathfrak{L}(S)\,B$.

Suppose now that $A\,\mathfrak{L}(S)\,B$, which is to say that for all $a \in A$ there exists a $b \in B$ such $\{a\}\,R\,\{b\}$. Because R is a bimodule this means that for all $a \in A$, $\{a\}\,R\,B$. But $A = \bigcup_{a\in A}\{a\}$, and therefore $A\,R\,B$, as R is left-disjunctive.

To show that this functor is dense we define a functor At : **CABARel** $\longrightarrow$ **Rel** which restricts relations to atoms:

$$\mathsf{At}(\mathcal{L}) \stackrel{\text{def}}{=} \{x \in \mathcal{L} \mid x \text{ is an atom of } \mathcal{L}\} \qquad\qquad x\,\mathsf{At}(R)\,y \stackrel{\text{def}}{\equiv} x\,R\,y$$

This operation restricts every relation to its atoms; it is evidently a functor.

Lemma 6. *The functor* $\mathfrak{L} : \mathbf{Rel} \longrightarrow \mathbf{CABARel}$ *is dense.*

Proof. Given a CABA $\mathcal{L}$ we define $R_{\mathcal{L}} : \mathcal{L} \nrightarrow \mathcal{P}(\mathsf{At}(\mathcal{L}))$ in **CABARel** where $x \, R_{\mathcal{L}} \, X$ just if $x \sqsubseteq \bigsqcup X$. Conversely, we define $R_{\mathcal{L}}^{-1} : \mathcal{P}(\mathsf{At}(\mathcal{L})) \nrightarrow \mathcal{L}$ by setting $X \, R_{\mathcal{L}}^{-1} \, x$ just if $\bigsqcup X \sqsubseteq x$. These relations are evidently directionally atomic and inverses of each other, as they compose to $\sqsubseteq$.

Theorem 2. $\mathbf{Rel} \simeq \mathbf{CABARel}$.

It is not difficult to show that the pseudo-inverse of $\mathfrak{L} : \mathbf{Rel} \longrightarrow \mathbf{CABARel}$ is actually the 'atoms' functor $\mathsf{At} : \mathbf{CABARel} \longrightarrow \mathbf{Rel}$ defined above.

Of course, Theorem 2 is not explicitly a duality; to make it one we have to compose it with the formal duality $(-)^{\dagger} : \mathbf{Rel}^{\mathsf{op}} \simeq \mathbf{Rel}$ [23, §2.3], obtaining a

Theorem 3 (Relational Tarski duality). $\mathbf{Rel}^{\mathsf{op}} \simeq \mathbf{CABARel}$.

It is possible to show that this duality is an 'extension' of the usual Tarski duality, in the sense that there is a commutative diagram of functors

$$\begin{tikzcd} \mathbf{Set}^{\mathsf{op}} \arrow[r, hook, "\mathrm{Gr}^{\mathsf{op}}"] \arrow[d, "\mathcal{P}"', "\simeq"] & \mathbf{Rel}^{\mathsf{op}} \arrow[d, "\simeq"', "\mathfrak{L}\circ(-)^{\dagger}"] \\ \mathbf{CABA} \arrow[r, hook, "j"'] & \mathbf{CABARel} \end{tikzcd} \tag{1}$$

where Gr and j are faithful and injective-on-objects.

We define j by taking each CABA $\mathcal{B}$ to itself, and each complete Boolean homomorphism $f^* : \mathcal{B} \to \mathcal{B}'$ to $j(f^*) : \mathcal{B} \nrightarrow \mathcal{B}'$ given by $b \, j(f^*) \, b' \overset{\text{def}}{\equiv} b \sqsubseteq f_!(b')$, where $f_!$ is the left adjoint of f^*.

Lemma 7. $j : \mathbf{CABA} \longrightarrow \mathbf{CABARel}$ *is a faithful functor.*

Proof. To begin we show that $j(f^*)$ is a morphism of **CABARel**. First, $j(f^*)$ is left-disjunctive: suppose that for each $i \in I$ we have that $x_i j(f^*) y$, i.e. $x_i \sqsubseteq f_!(y)$. Then $\bigsqcup_{i \in I} x_i \sqsubseteq f_!(y)$, and hence $\bigsqcup_{i \in I} x_i j(f^*) y$. Second, $j(f^*)$ is atomic-founded. Suppose $x \, j(f^*) \, y$, i.e. $x \sqsubseteq f_!(y)$, with x an atom. Write $y = \bigsqcup_{i \in I} a_i$ for a_i atoms. Then $f_!(y) = \bigsqcup_{i \in I} f_!(a_i)$, because $f_!$ is a left adjoint and preserves joins. As x is an atom, $x \sqsubseteq f_!(a_i)$ for a particular $i \in I$, so $x \, j(f^*) \, a_i$ for some $a_i \sqsubseteq y$. Thirdly, $j(f^*)$ is evidently a bimodule, by monotonicity of $f_!$.

j preserves identities: we have $x \, j(\mathsf{id}) \, y$ iff $x \sqsubseteq \mathsf{id}_!(y) = y$, viz. the identity in **CABARel**. j preserves composition: let $f^* : X \to Y$ and $g^* : Y \to Z$ be complete Boolean homomorphisms. Then $x \, j(g^* \circ f^*) \, z$ iff $x \sqsubseteq f_!(g_!(z))$ iff $x j(f^*) g_!(z)$. But by reflexivity $g_!(z) \sqsubseteq g_!(z)$, i.e. $g_!(z) j(g^*) z$, so $x (j(f^*) \, ; j(g^*)) z$. For the converse, suppose $x \, j(f^*) \, y \, j(g^*) \, z$, i.e. $x \sqsubseteq f_!(y)$ and $y \sqsubseteq g_!(z)$. By the monotonicity of $f_!$ we have $x \sqsubseteq f_!(y) \sqsubseteq f_!(g_!(z))$, therefore $z \, j(g^* \circ f^*) \, x$.

Finally, j is evidently faithful by a Yoneda-type argument.

We define Gr by taking each set X to itself, and each function $f : X \to Y$ to its *graph* $\mathrm{Gr}\,(f) : X \nrightarrow Y$, viz. $x \, \mathrm{Gr}\,(f) \, y \overset{\text{def}}{\equiv} (f(x) = y)$.

Lemma 8. *The functor* $Gr : \mathbf{Set} \longrightarrow \mathbf{Rel}$ *is faithful.*

Proof. Gr clearly preserves identity and composition. It is faithful because functions are entirely determined by their graph.

Finally, it is simple to calculate that (1) commutes.

Sketch of a formal system Having developed this duality we can now see how it can be used for relating formulae. Suppose we have a relation $R : X \to\!\!\!\!\to Y$. Under the relational Tarski duality this induces a relation $\mathfrak{L}(R) : \mathcal{P}(X) \to\!\!\!\!\to \mathcal{P}(Y)$ between predicates. Let us notate this relation as the judgment

$$\varphi \overset{R}{\searrow} \psi$$

where $\varphi \in \mathcal{P}(X)$ is a predicate over X and $\psi \in \mathcal{P}(Y)$ over Y intuitively says that *every state of* X *that satisfies* φ *is* R*-related to some state of* Y *that satisfies* ψ. The fact $\mathfrak{L}(R)$ is a bimodule means that the rule

$$\frac{\varphi' \vdash \varphi \qquad \varphi \overset{R}{\searrow} \psi \qquad \psi \vdash \psi}{\varphi' \overset{R}{\searrow} \psi'}$$

is sound. This is reminiscent of the *consequence rule* of Hoare logic [15].

The fact $\mathfrak{L}(R)$ is left-disjunctive means that the rule

$$\frac{\varphi_1 \overset{R}{\searrow} \psi \qquad \varphi_2 \overset{R}{\searrow} \psi}{(\varphi_1 \vee \varphi_2) \overset{R}{\searrow} \psi}$$

is sound. This allows us reasoning by cases on the left. The final characteristic property of $\mathfrak{L}(R)$, namely its atomic-foundedness, would require introducing judgements that capture atomicity of predicates, in the style of Abramsky [3].

Finally, the fact $\mathfrak{L}$ is a functor can be formally expressed by the rules

$$\frac{\varphi \overset{R}{\searrow} \psi \qquad \psi \overset{S}{\searrow} \chi}{\varphi \overset{R;S}{\searrow} \chi} \qquad\qquad \frac{\phi \vdash \psi}{\varphi \overset{\mathsf{id}}{\searrow} \psi} \qquad\qquad \frac{\varphi \overset{\mathsf{id}}{\searrow} \psi}{\phi \vdash \psi}$$

5 A Modal Relational Duality

In this section we will extend the relational Tarski duality to classical modal logic. The basis for this is *Thomason duality* [37] [23, §2.4], which uniquely associates *Kripke frames* with CABAs equipped with a modal operator.

A *Kripke frame* (X, R) (also called a *transition system*) is a set X equipped with a relation $R : X \to\!\!\!\!\to X$. The intuition is that X is a set of worlds, and the relation R encodes *transitions* from one world to another. We will often write

$x \to_R y$ instead of $x_1 \, R \, x_2$. A morphism of Kripke frames $f : (X, R) \to (Y, S)$ is a function $f : X \to Y$ which preserves transitions, in the sense that $x_1 \to_R x_2$ implies $f(x_1) \to_S f(x_2)$. Kripke frames and their morphisms a category **Frm**.

The relation R can also be seen as a map $\lambda R : X \to \mathcal{P}(X)$ by currying. It is then the case that there is a unique join-preserving function $\blacklozenge_R : \mathcal{P}(X) \to \mathcal{P}(X)$ that makes the following diagram commute, where $\mathbf{y} : X \to \mathcal{P}(X)$ maps every element $x \in X$ to the singleton set $\{x\}$.[5]

$$\begin{tikzcd} X \arrow[r, "\mathbf{y}"] \arrow[dr, "\lambda R"'] & \mathcal{P}(X) \arrow[d, dashed, bend right, "\blacklozenge_R"'] \arrow[d, phantom, "\dashv"] \\ & \mathcal{P}(X) \arrow[u, dotted, bend right, "\Box_R"'] \end{tikzcd}$$

As $\blacklozenge_R$ preserves joins and $\mathcal{P}(X)$ is a complete lattice, the adjoint functor theorem implies that it has a right adjoint $\Box_R : \mathcal{P}(X) \to \mathcal{P}(X)$, which preserves all meets. These maps are explicitly given by

$$\begin{aligned} \blacklozenge_R(A) &= \{w \in X \mid \exists v \in A.\ v \, R \, w\} \\ \Box_R(A) &= \{w \in X \mid \forall v \in X.\ w \, R \, v \implies v \in A\} \end{aligned}$$

Having one of these three pieces of data (the relation R; a join-preserving map $\blacklozenge_R$; a meet-preserving map $\Box_R$) uniquely determines the other two [21].

It is possible to define morphisms of frames purely in terms of operators:

Lemma 9. *A function $f : X \to Y$ is a morphism of frames $f : (X, R) \to (Y, S)$ if and only if $f^* \circ \Box_S \subseteq \Box_R \circ f^*$.*

Let **CABAO** be the category whose objects $(\mathcal{B}, \Box_{\mathcal{B}})$ are CABAs $\mathcal{B}$ equipped with an *operator* $\Box : \mathcal{B} \to \mathcal{B}$ that preserves all meets,[6] and morphisms $f^* : (\mathcal{B}, \Box_{\mathcal{B}}) \to (\mathcal{B}', \Box_{\mathcal{B}'})$ are complete Boolean homomorphisms $f^* : \mathcal{B} \to \mathcal{B}'$ that satisfy $f^* \circ \Box_{\mathcal{B}} \sqsubseteq \Box_{\mathcal{B}'} \circ f^*$ (with the pointwise order). This yields a modal duality

Theorem 4 (Weak Thomason duality). $\mathbf{Frm}^{\mathsf{op}} \simeq \mathbf{CABAO}$.

The maps between CABAOs preserve the Boolean structure, but they only preserve $\Box$ weakly. To strengthen this we need the notion of an *open map.*

Definition 6. *A morphism of frames $f : (X, R) \to (Y, S)$ is* open *just if $f(x) S y'$ implies that there exists a $x' \in X$ with $x \, R \, x'$ and $f(x') = y$.*

We let $\mathbf{Frm}_{\text{open}}$ be the wide subcategory of **Frm** whose morphisms are open. It is easy to show that such an f is open iff $f^* \circ \Box_S = \Box_R \circ f^*$. Then, letting $\mathbf{CABAO}_{\text{open}}$ be the wide subcategory of **CABAO** whose morphisms f^* preserve $\Box$ (i.e. $f^* \circ \Box = \Box \circ f^*$) we obtain a refinement of Theorem 4:

[5] This is also known as the *left Kan extension* of λR along the Yoneda embedding.
[6] Or, equivalently, a join-preserving operator $\blacklozenge$

Theorem 5 (Thomason duality). $\mathbf{Frm}^{\mathsf{op}}_{open} \simeq \mathbf{CABAO}_{open}$.

To obtain a relational version of this duality we must relax the functionality of morphisms of Kripke frames. To achieve that, notice that the notion of an open map is precisely a *functional bisimulation* [33, §3.2].

Definition 7 (Simulations, Cosimulations, and Bisimulations). *Let (X, R) and (Y, S) be Kripke frames, and let $Q : X \nrightarrow Y$ be a relation.*

1. *Q is a* simulation *whenever $x\,Q\,y$ and $x\,R\,x'$ we have a $y' \in Y$ with $y\,S\,y'$ and $x'\,Q\,y'$. In that case we write $Q : (X, R) \nrightarrow (Y, S)$.*
2. *Q is a* cosimulation *whenever $Q^\dagger$ is a simulation.*
3. *Q is a* bisimulation *if it is both a simulation and a cosimulation.*

We may illustrate the definitions of simulation and cosimulation pictorially:

$$\begin{tikzcd} x' \arrow[r, dashed, "Q"] & \exists y' \\ x \arrow[u, "R"] \arrow[r, "Q"] & y \arrow[u, dotted, "S"'] \end{tikzcd} \qquad \begin{tikzcd} \exists x' \arrow[r, dashed, "Q"] & y' \\ x \arrow[u, dotted, "R"] \arrow[r, "Q"] & y \arrow[u, "S"'] \end{tikzcd}$$

Considering an open map as a relation, the fact it preserves transitions means it is a simulation (i.e. it satisfies the 'forth' condition). The fact it is open means it is a cosimulation (i.e. it satisfies the 'back' condition).

It is easy to see that simulations and cosimulations are closed under relational composition, and that the identity relation is a bisimulation. We thus obtain two categories **FrmSim** and **FrmBisim** with Kripke frames as objects, and simulations and bisimulations as morphisms respectively. Noting that the opposite of a simulation is a cosimulation, and vice versa, there is an obvious formal duality $(-)^\dagger : \mathbf{FrmBisim}^{\mathsf{op}} \simeq \mathbf{FrmBisim}$, meaning **FrmBisim** is also a dagger category.

It is possible to characterise simulations purely in terms of the lower relation.

Lemma 10. *A relation $Q : X \nrightarrow Y$ is a simulation $Q : (X, R) \nrightarrow (Y, S)$ if and only if $A\,\mathfrak{L}(Q)\,B$ implies $\blacklozenge A\,\mathfrak{L}(Q)\,\blacklozenge B$ for any $A \subseteq X$ and $B \subseteq Y$.*

Proof. Suppose that $Q : (X, R) \nrightarrow (Y, S)$ is a simulation and that $A\,\mathfrak{L}(Q)\,B$. If $x \in \blacklozenge A$, then there exists some $a \in A$ with $a \to_R x$. We know that there exists some $b \in B$ for which $a\,Q\,b$, so by simulation there must exist some $y \in Y$ such that $b \to_S y$, so $y \in \blacklozenge B$, and $x\,Q\,y$.

For the converse, suppose that if $A\,\mathfrak{L}(Q)\,B$ then $\blacklozenge A\,\mathfrak{L}(Q)\,\blacklozenge B$, and that $x\,Q\,y$. Then we know that $\{x\}\ \mathfrak{L}(Q)\ \{y\}$ and therefore $\blacklozenge\{x\}\ \mathfrak{L}(Q)\ \blacklozenge\{y\}$. So, for any element $x' \in X$ with $x \to_R x'$ we know that $x' \in \blacklozenge\{x\}$, therefore there exists some $y' \in \blacklozenge\{y\}$—that is, $y' \in Y$ with $y \to_S y'$—for which $x'\,Q\,y'$.

It is also possible to state this property in what we call a 'half-adjoint' style.

Lemma 11. *Let (X, R) and (S, Q) be Kripke frames, and $Q : X \nrightarrow Y$. Then Q is a simulation $(X, R) \nrightarrow (Y, S)$ iff $A\,\mathfrak{L}(Q)\,\Box B$ implies $\blacklozenge A\,\mathfrak{L}(Q)\,B$.*

Proof. For the forwards direction suppose that $A\ \mathfrak{L}(Q)\ \Box B$. Because Q is a simulation we have $\blacklozenge A\ \mathfrak{L}(Q)\ \blacklozenge\Box B$ by Lemma 10. But $\blacklozenge \dashv \Box$ so $\blacklozenge\Box B \subseteq B$, and hence $(\blacklozenge A)\ \mathfrak{L}(Q)\ B$ as $\mathfrak{L}(Q)$ is a bimodule.

For the backwards direction suppose $A\ \mathfrak{L}(Q)\ B$. Notice that as $\blacklozenge \dashv \Box$ we have $B \subseteq \Box\blacklozenge B$. As $\mathfrak{L}(Q)$ is a bimodule we know that $A\ \mathfrak{L}(Q)\ (\Box\blacklozenge B)$. Then by assumption $\blacklozenge A\ \mathfrak{L}(Q)\ \blacklozenge B$, which by Lemma 10 implies that Q is a simulation.

Define a *simulatory relation* $Q : (X, \Box_R) \twoheadrightarrow (Y, \Box_S)$ between CABAOs to be a directionally atomic $Q : X \twoheadrightarrow Y$ for which the condition of Lemma 11 holds, i.e. $A\ Q\ \Box_S B$ implies $\blacklozenge_R A\ Q\ B$.

Proposition 3. *If $R : (X, \Box_X) \twoheadrightarrow (Y, \Box_Y)$ and $S : (Y, \Box_Y) \twoheadrightarrow (Z, \Box_Z)$ are simulatory relations, then their relational composition is a simulatory relation $R\,;S : (X, \Box_X) \twoheadrightarrow (Z, \Box_Z)$.*

Proof. We already know that $R;S$ is directionally atomic. Suppose that $A(R\,;S)$ $\Box C$, i.e. there exists some B with $ARBS\Box C$. Then by $\blacklozenge \dashv \Box$ we have $B \subseteq \Box\blacklozenge B$, therefore $A\ R\ \Box\blacklozenge B$ as R is a bimodule. By assumption $\blacklozenge A\ R\ \blacklozenge B$. Also by assumption $\blacklozenge B\ S\ C$, whence $\blacklozenge A\ (R\,;S)\ C$.

Define **CABAOSim** to have CABAs with operators $(\mathcal{B}, \Box_{\mathcal{B}})$ as objects and simulatory relations $Q : (X, \Box_R) \twoheadrightarrow (Y, \Box_S)$ as morphisms. We will often write these morphisms as simply $Q : X \twoheadrightarrow Y$ and omit subscripts. Using the preceding lemma it is easy to check that this is a category, with $\sqsubseteq$ as an identity morphism.

Lemma 12. *$\mathfrak{L}$ is a functor* **FrmSim** $\longrightarrow$ **CABAOSim**.

Proof. $\mathfrak{L}(Q)$ is directionally atomic by Lemma 3. Moreover, $\mathfrak{L}(Q)$ is a simulatory relation by Lemma 11. Simulatory relations compose by Proposition 3.

Lemma 13. $\mathfrak{L}$: **FrmSim** $\longrightarrow$ **CABAOSim** *is faithful.*

Proof. Immediate from Lemma 4.

Lemma 14. $\mathfrak{L}$: **FrmSim** $\longrightarrow$ **CABAOSim** *is full. In other words, every simulatory relation $Q : (\mathcal{P}(X), \Box_R) \twoheadrightarrow (\mathcal{P}(Y), \Box_S)$ is $\mathfrak{L}(T)$ for some simulation $T : (X, R) \twoheadrightarrow (Y, S)$.*

Proof. By Lemma 5 we know that $Q = \mathfrak{L}(S_Q)$ where $x\ S_Q\ y$ iff $\{x\}\ Q\ \{y\}$. By assumption $A\ \mathfrak{L}(S_Q)\ \Box B$ implies $\blacklozenge A\ \mathfrak{L}(S_Q)\ B$, so by Lemma 11 we know that S_Q is a simulation .

We can use the same definition for the functor At : **CABAOSim** $\longrightarrow$ **FrmSim** as before, which simply restricts relations to atoms. By Lemma 14 it is clearly a functor.

Lemma 15. $\mathfrak{L}$: **FrmSim** $\longrightarrow$ **CABAOSim** *is dense.*

Proof. Following the proof of Lemma 6, given a **CABAO** $\mathcal{B}$, define the relations $R_{\mathcal{B}} : \mathcal{B} \to\!\!\!\!\to \mathcal{P}(\mathsf{At}(\mathcal{B}))$ and $R_{\mathcal{B}}^{-1} : \mathcal{P}(\mathsf{At}(\mathcal{B})) \to\!\!\!\!\to \mathcal{B}$ by $x \, R_{\mathcal{B}} \, X$ just if $x \sqsubseteq \bigsqcup X$ and $X \, R_{\mathcal{B}}^{-1} \, c$ just if $\bigsqcup X \sqsubseteq x$ respectively. As in the proof of Lemma 6 these are both directionally atomic and evidently inverses to each other. Thus, it suffices to show that they are simulatory.

Suppose that $x \sqsubseteq \bigsqcup \Box X$, then

$$x \sqsubseteq \bigsqcup \Box X \sqsubseteq \Box \blacklozenge \bigsqcup \Box X = \Box \bigsqcup \blacklozenge \Box X \sqsubseteq \Box \bigsqcup X$$

by properties of the adjunction $\blacklozenge \dashv \Box$. Hence $\blacklozenge x \sqsubseteq \bigsqcup X$, and $R_{\mathcal{B}}$ is simulatory.

Similarly, if $\bigsqcup X \sqsubseteq \Box x$, then $\blacklozenge \bigsqcup X \sqsubseteq x$ and therefore $\bigsqcup \blacklozenge X \sqsubseteq x$, so $R_{\mathcal{B}}^{-1}$ is simulatory.

In summary, we obtain an equivalence

Theorem 6. **FrmSim** $\simeq$ **CABAOSim**.

Our final objective is to restrict this equivalence by analogy to the restriction of $\mathbf{Frm}^{\mathsf{op}} \simeq \mathbf{CABAO}$ to $\mathbf{Frm}^{\mathsf{op}}_{\text{open}} \simeq \mathbf{CABAO}_{\text{open}}$, where the maps in $\mathbf{Frm}^{\mathsf{op}}_{\text{open}}$ are open, and correspondingly the maps in $\mathbf{CABAO}_{\text{open}}$ preserve all logical connectives. In our relational duality the morphisms are no longer functional, but merely simulations. Thus, openness will turn them into *bisimulations.*

Unfortunately, expressing bisimilarity using our current vocabulary is not evidently possible. For that we will need the following 'dual' relational lifting.

Definition 8 (Upper Relation). *Given a relation $R : X \to\!\!\!\!\to Y$, the associated upper relation $\mathfrak{U}(R) : \mathcal{P}(X) \to\!\!\!\!\to \mathcal{P}(Y)$ is defined by*

$$S \, \mathfrak{U}(R) \, T \stackrel{\text{def}}{\equiv} \forall s \in S. \; \exists t \in T. \; s \, R \, t.$$

It is easy to see that $\mathfrak{U}(R) = \mathfrak{L}(R^{\dagger})^{\dagger}$ for any $R : X \to\!\!\!\!\to Y$. This makes the theory perfectly dual, in the sense that $\mathfrak{U}(R)$ is *opdirectionally atomic* (i.e. right-disjunctive, atomic op-founded, and an opbimodule). To avoid this proliferation of concepts we will simply work with the opposite relation $\mathfrak{L}(R^{\dagger})$. Somewhat surprisingly, we can construct this relation directly on relations between CABAs.

Definition 9. *For a relation $R : \mathcal{B} \to\!\!\!\!\to \mathcal{B}'$ between CABAs we define its* variant relation *$R^{\bullet} : \mathcal{B}' \to\!\!\!\!\to \mathcal{B}$ to be*

$$b' \, R^{\bullet} \, b \stackrel{\text{def}}{\equiv} \forall \text{atom } a' \sqsubseteq b'. \; \exists \text{atom } a \sqsubseteq b. \; a \, R \, a'.$$

Lemma 16. *For a relation $R : \mathcal{B} \to\!\!\!\!\to \mathcal{B}'$ between CABAs, the variant $R^{\bullet} : \mathcal{B}' \to\!\!\!\!\to \mathcal{B}$ is directionally atomic.*

Proof. First, we prove it is left-disjunctive. Suppose that $b'_i \, R^{\bullet} \, b$ for all $i \in I$, some indexing set. Then see that for any atom $a' \sqsubseteq \bigsqcup_i b'_i$, there exists some $i \in I$ for which $a' \sqsubseteq b'_i$, therefore there exists an atom $a \sqsubseteq b_i$ for which $a \, R \, a'$. In particular, $(\bigsqcup_i b'_i) \, R^{\bullet} \, b$.

Second, we prove it is atomic founded. Suppose that b' is an atom and $b' \, R^\bullet \, b$. See that $b' \sqsubseteq b$, so there exists an atom $a \sqsubseteq b$ such that $a \, R \, b'$. In particular $b' \, R^\bullet \, a$, as $a \sqsubseteq a$.

Finally, we prove it is a bimodule. Suppose that $b' \, R^\bullet \, b$, and $c' \sqsubseteq b'$ and $b \sqsubseteq c$. Then for any atom $a' \sqsubseteq c' \sqsubseteq b'$, there must exist some atom $a \sqsubseteq b \sqsubseteq c$ for which $a \, R \, a'$.

Lemma 17. *The variant preserves composition contravariantly, that is, for relations $R : X \twoheadrightarrow Y$ and $S : Y \twoheadrightarrow Z$ between CABAs, $(R \,;\, S)^\bullet \equiv S^\bullet \,;\, R^\bullet$.*

Proof. First, suppose $z \, (R \,;\, S)^\bullet \, x$. Then, for any atom $c \sqsubseteq z$ we know that there exists some atom $a \sqsubseteq x$ such that $a \, (R \,;\, S) \, c$. In particular, because R is atom founded and S is a bimodule, there must exist some atom $b \in Y$ such that $a \, R \, b \, S \, c$. Define

$$b_c \stackrel{\text{def}}{=} \bigsqcup \{\text{atom } b \in Y \mid b \, S \, c \wedge \exists \text{atom } a \sqsubseteq x.\ a \, R \, b\} \qquad y \stackrel{\text{def}}{=} \bigsqcup_{\text{atom } c \sqsubseteq z} b_c,$$

Then for any atom $c \sqsubseteq z$, by definition there exists an atom $b \sqsubseteq y$ for which $b \, S \, c$, and for that atom b there is some atom $a \sqsubseteq x$ such that $a R b$. So $z \, S^\bullet \, y \, R^\bullet \, x$

Conversely, suppose $z \, S^\bullet \, y \, R^\bullet \, x$ for some $y \in Y$. Take some atom $c \sqsubseteq z$, then there exists an atom $b \sqsubseteq y$ for which $b \, S \, c$, and for that b there exists some atom $a \sqsubseteq x$ for which $a \, R \, b$, so $a \, (R \,;\, S) \, c$. In particular, $z \, (R \,;\, S)^\bullet \, x$.

Lemma 18. $R^{\bullet\bullet} = R$ *for any directionally atomic $R : \mathcal{B} \twoheadrightarrow \mathcal{B}'$.*

Proof. First, notice that $b \, R^{\bullet\bullet} \, b'$ just if for any atom $a \sqsubseteq b$ there exists an atom $a' \sqsubseteq b'$ for which $a \, R \, a'$, simply by the atomicity of a and a'. That is, $a' \, R^\bullet \, a$ just if for any atom $x' \sqsubseteq a'$ there exists an atom $x \sqsubseteq a$ for which $x \, R \, x'$, but by atomicity we know that $x' = a'$ and $x = a$.

For the forward direction, suppose that $b \, R^{\bullet\bullet} \, b'$, so $\forall$atom $a \sqsubseteq b$. $\exists$atom $a' \sqsubseteq b'$. $a \, R \, a'$. But because R is a bimodule, we know that $\forall$atom $a \sqsubseteq b$. $a \, R \, b'$. Furthermore, because it is left-disjunctive and by the atomicity of $\mathcal{B}$ we know that $b R b'$. For the backward direction, suppose that $b R b'$. Take any atom $a \sqsubseteq b$, then by bimodule we know that $a \, R \, b'$. By atom-founded, we know that there exists an atom $a' \sqsubseteq b'$ such that $a \, R \, b'$. Therefore $b \, R^{\bullet\bullet} \, b'$.

This acts in the expected way between lifted relations.

Lemma 19. $\mathfrak{L}(R)^\bullet = \mathfrak{L}(R^\dagger)$ *for any $R : X \twoheadrightarrow Y$, and hence $\mathfrak{U}(R) = \mathfrak{L}(R)^{\bullet\dagger}$.*

Proof. Let $R : X \twoheadrightarrow Y$ be a relation between sets. Then for subsets $T \subseteq Y$ and $S \subseteq X$ we have $T \, \mathfrak{L}(R)^\bullet \, S$ just if $\forall s \in S.\ \exists t \in T.\ t \, R \, s$. Note that this is exactly the same as $T \, \mathfrak{L}(R^\dagger) \, S$.

Like with simulations and Lemma 10, it is possible to characterise cosimulations purely in terms of the variant relation.

Lemma 20. *A relation $Q : X \twoheadrightarrow Y$ is a cosimulation $Q : (X, R) \twoheadrightarrow (Y, S)$ if and only if $B \, \mathfrak{L}(Q)^\bullet \, A$ implies $\blacklozenge B \, \mathfrak{L}(Q)^\bullet \, \blacklozenge A$ for any $B \subseteq Y$ and $A \subseteq X$.*

Proof. By noticing that R is a cosimulation iff $R^\dagger$ is a simulation, and then combining Lemmas 10 and 19.

We can also characterise this condition as a 'half-adjunction' (cf Lemma 11).

Lemma 21. *Let (X, R) and (S, Q) be Kripke frames, and $Q : X \twoheadrightarrow Y$. Then Q is a cosimulation $(X, R) \twoheadrightarrow (Y, S)$ iff $B\ \mathfrak{L}(Q)^\bullet\ \Box A$ implies $\blacklozenge B\ \mathfrak{L}(Q)^\bullet\ A$.*

The proof is completely dual to Lemma 11.

Define a *cosimulatory relation* $Q : (X, \Box_R) \twoheadrightarrow (Y, \Box_S)$ between CABAOs to be a directionally atomic $Q : X \twoheadrightarrow Y$ for which the condition of lemma 21 holds, i.e. $B\ Q^\bullet\ \Box_R A$ implies $\blacklozenge_S B\ Q^\bullet\ A$.

The following proposition is analogous to Proposition 3; the proof is dual.

Proposition 4. *If $R : (X, \Box_X) \twoheadrightarrow (Y, \Box_Y)$ and $S : (Y, \Box_Y) \twoheadrightarrow (Z, \Box_Z)$ are cosimulatory relations, then their relational composition is a cosimulatory relation $R\,;S : (X, \Box_X) \twoheadrightarrow (Z, \Box_Z)$.*

Call such a relation *bisimulatory* if it is both simulatory and cosimulatory. Define **CABAOBisim** to have CABAs with operators $(\mathcal{B}, \Box_\mathcal{B})$ as objects and bisimulatory relations $Q : (X, \Box_R) \twoheadrightarrow (Y, \Box_S)$ as morphisms. This is again a category with $\sqsubseteq$ as an identity morphism.

Lemma 22. *$\mathfrak{L}$ is a functor* **FrmBisim** $\longrightarrow$ **CABAOBisim**.

Proof. $\mathfrak{L}(Q)$ is directionally atomic by Lemma 3. Moreover, $\mathfrak{L}(Q)$ is bisimulatory by Lemmas 11 and 21. Bisimulatory relations compose by Propositions 3 and 4.

Lemma 23. $\mathfrak{L}$: **FrmBisim** $\longrightarrow$ **CABAOBisim** *is faithful.*

Proof. Immediate from Lemma 4.

Lemma 24. $\mathfrak{L}$: **FrmBisim** $\longrightarrow$ **CABAOBisim** *is full. In other words, every simulatory relation $Q : (\mathcal{P}(X), \Box_R) \twoheadrightarrow (\mathcal{P}(Y), \Box_S)$ is $\mathfrak{L}(T)$ for some bisimulation $T : (X, R) \twoheadrightarrow (Y, S)$.*

Proof. By Lemma 5 we know that $Q = \mathfrak{L}(S_Q)$ where $x\ S_Q\ y$ iff $\{x\}\ Q\ \{y\}$. By assumption $A\ \mathfrak{L}(S_Q)\ \Box B$ implies $\blacklozenge A\ \mathfrak{L}(S_Q)\ B$, so by Lemma 11 we know that S_Q is a simulation between frames.

Lemma 25. $\mathfrak{L}$: **FrmBisim** $\longrightarrow$ **CABAOBisim** *is dense.*

Proof. We augment the proof of Lemma 15 by showing that both $R_\mathcal{B} : \mathcal{B} \twoheadrightarrow \mathcal{P}(\mathsf{At}(\mathcal{B}))$ and $R_\mathcal{B}^{-1} : \mathcal{P}(\mathsf{At}(\mathcal{B})) \twoheadrightarrow \mathcal{B}$ given by $x\ R_\mathcal{B}\ X$ just if $x \sqsubseteq \bigsqcup X$ and $X\ R_\mathcal{B}^{-1}\ c$ just if $\bigsqcup X \sqsubseteq x$ respectively are also cosimulatory.

First, see that $X\, {R_\mathcal{B}}^\bullet\, x$ just if for any atom $A \sqsubseteq X$ there exists an atom $a \sqsubseteq x$ for which $a \sqsubseteq \bigsqcup A$. But by atomicity, this is the same as $a = A$, so $X\ {R_\mathcal{B}}^\bullet\ x$ just if $X \sqsubseteq x$. Hence it is obvious that $X \sqsubseteq \Box x$ implies $\blacklozenge X \sqsubseteq x$, and $R_\mathcal{B}$ is cosimulatory.

Second, see that $x\ {R_\mathcal{B}^{-1}}^\bullet\ X$ just if for any atom $a \sqsubseteq x$ there exists an atom $A \subseteq X$ for which $\bigsqcup A \sqsubseteq a$. But by atomicity this is just $A = \{a\}$, so $x\ {R_\mathcal{B}^{-1}}^\bullet\ X$ just if $x \sqsubseteq X$. Hence it is obvious that $x \sqsubseteq \Box X$ implies $\blacklozenge x \sqsubseteq X$, and $R_\mathcal{B}^{-1}$ is cosimulatory.

We thus obtain the following equivalence.

Theorem 7. $\mathbf{FrmBisim} \simeq \mathbf{CABAOBisim}$.

As before, this is not explicitly a duality, but we can obtain one from the formal duality $(-)^\dagger : \mathbf{FrmBisim}^{\mathsf{op}} \simeq \mathbf{FrmBisim}$.

Theorem 8 (Relational Thomason duality). $\mathbf{FrmBisim}^{\mathsf{op}} \simeq \mathbf{CABAOBisim}$.

We can show that Theorem 8 is an extension of the Thomason duality; as with (1), there is a commutative diagram of functors

$$\begin{tikzcd}
\mathbf{Frm}^{\mathsf{op}}_{\text{open}} \arrow[r, hook, "\mathrm{Gr}^{\mathsf{op}}"] \arrow[d, "\mathcal{P}"', "\simeq"] & \mathbf{FrmBisim}^{\mathsf{op}} \arrow[d, "\simeq"', "\mathfrak{L}\circ(-)^\dagger"] \\
\mathbf{CABAO}_{\text{open}} \arrow[r, hook, "j"'] & \mathbf{CABAOBisim}
\end{tikzcd} \tag{2}$$

where Gr and j are faithful and injective-on-objects.

We define j as before, by taking each CABA-operator pair $(\mathcal{B}, \Box_{\mathcal{B}})$ to itself, and each complete Boolean homomorphism $f^* : \mathcal{B} \to \mathcal{B}'$ to $j(f^*) : \mathcal{B} \twoheadrightarrow \mathcal{B}'$ given by $b\, j(f^*)\, b' \overset{\text{def}}{\equiv} b \sqsubseteq f_!(b')$.

Lemma 26. *$j : \mathbf{CABAO}_{open} \longrightarrow \mathbf{CABAOBisim}$ is a faithful functor.*

Proof. By extension of lemma 7, we know that $j(f^*)$ is a directionally atomic relation, and that j preserves identities and composition, and that it is faithful. It remains only to show that $j(f^*)$ is bisimulatory.

First, to show that $j(f^*)$ is simulatory, suppose that $x\, j(f^*)\, (\Box y)$, that is, $x \sqsubseteq f_!(\Box y)$. Then

$$x \sqsubseteq f_!(\Box y) \sqsubseteq \Box\blacklozenge f_!(\Box y) \sqsubseteq \Box f_!(\blacklozenge\Box y) \sqsubseteq \Box f_!(y)$$

by the properties of the adjunction $\blacklozenge \dashv \Box$. Hence $\blacklozenge x \sqsubseteq f_!(y)$, so $j(f^*)$ is simulatory.

Second, to show that $j(f^*)$ is cosimulatory, notice first that $b\, j(f^*)\, a$ just if, for any atom $y \sqsubseteq b$ there exists some atom $x \sqsubseteq a$ for which $x \sqsubseteq f_!(y)$, which is the same as $x = f_!(y)$ by the properties of atoms and recalling that $f_!$ preserves atoms. Therefore, $b\, j(f^*)\, a$ just if $b \sqsubseteq f^*(a)$. Hence it is easy to see that, if $b \sqsubseteq f^*(\Box a) = \Box f^*(a)$, then $\blacklozenge b \sqsubseteq f^*(a)$, and $j(f^*)$ is cosimulatory.

As usual, Gr takes each frame (X, R) to itself and each open map $f : (X, R) \to (Y, S)$ to the graph $\mathrm{Gr}(f) : X \twoheadrightarrow Y$ defined by $x\, \mathrm{Gr}(f)\, y \overset{\text{def}}{\equiv} f(x) = y$.

Lemma 27. *The functor Gr : $\mathbf{Frm}_{open} \longrightarrow \mathbf{FrmBisim}$ is faithful.*

Proof. Gr clearly takes open maps to bisimulations, preserves identity and composition. It is faithful because any function is determined entirely by its graph.

Finally, it is simple to calculate that (2) commutes.

Sketch of a formal system Recall the formal system of the judgment $\overset{Q}{\searrow}$ for a relation $Q : X \nrightarrow Y$ that we sketched in §4. Now assume that (X, R) and (Y, S) are Kripke frames. We can extend this system to a modal logic where the logic over X and Y have corresponding modalities $\blacklozenge$ and $\square$. If Q is a simulation then by Lemmas 10 and 11 we can extend the system with the rules

$$\frac{\varphi \overset{R}{\searrow} \psi}{\blacklozenge\varphi \overset{R}{\searrow} \blacklozenge\psi} \qquad\qquad \frac{\varphi \overset{R}{\searrow} \square\psi}{\blacklozenge\varphi \overset{R}{\searrow} \psi}$$

If R is a cosimulation we can use Lemmas 20 and 21 to add the rules

$$\frac{\varphi \overset{R^\dagger}{\searrow} \psi}{\blacklozenge\varphi \overset{R^\dagger}{\searrow} \blacklozenge\psi} \qquad\qquad \frac{\varphi \overset{R^\dagger}{\searrow} \square\psi}{\blacklozenge\varphi \overset{R^\dagger}{\searrow} \psi}$$

6 Related work

A few relational dualities have been previously described in the literature. However, most are of the form $\mathcal{C}^{\mathsf{op}} \simeq \mathcal{D}$ where $\mathcal{C}$ is a category whose morphisms are relations, whereas $\mathcal{D}$ is a category of algebras with some form of *hemimorphism*, i.e. a morphism preserving most—but not all!–the logical structure. The earliest duality of the form, which we mentioned in §4 is $\mathbf{Rel} \simeq \mathbf{CABA}_{\bigvee}$. This duality has been rediscovered multiple times, but is likely due to Jonsson and Tarski [20]. Kishida [23, §2.3] argues that this extends to a '2-duality,' as both of these are (simple) 2-categories. Halmos [11] extended this to a Stone duality between Stone spaces with continuous relations on the one hand, and Boolean algebras with *hemimorphisms* (i.e. functions preserving only $\perp$ and $\vee$). Cignoli, Lafalce, and Petrovich [7] do something similar for Priestly spaces and continuous monotone relations. Hofmann and Nora [16] have proposed a general framework for such dualities, obtaining the relational side as the Kleisli category of a suitable monad. Hofmann and Nora [17] extend such dualities from order theory to metric structures, and quantale-enriched categories on the other side.

Kurz, Moshier, and Jung [27] present dualities that are much closer to the flavour of the relational dualities we consider in this paper. They achieve this by working in an order-enriched setting, where relations can be presented as both spans and cospans in a 2-categorical manner. They use this to extend (well-behaved) dualities to dualities between categories of relations, and even adjunctions of categories to adjunctions between framed bicategories [35]. For example, if their recipe is applied to $\mathbf{Pos}$, it lifts a bimodule $R : X \nrightarrow Y$ to the relation $2^R : [X, 2] \nrightarrow [Y, 2]$ between upper sets given by $A\, 2^R\, B$ just if for all $x \in A$ and $x\, R\, y$ implies $y \in B$. Hence, the type of lifting they obtain is very different, and it is not evident that it is related to bisimulations.

Birkmann, Urbat, and Milius [4] present extensions of categorical dualities through monoidal adjunctions and apply them to algebraic language theories.

As a corollary they obtain some relational dualities, e.g. between well-behaved relations of profinite ordered monoids and natural morphisms of residuation algebras.

References

1. Abramsky, S.: Domain Theory and the Logic of Observable Properties. PhD thesis, Queen Mary College, University of London (1987). https://www.cs.ox.ac.uk/files/350/thesis.pdf.
2. Abramsky, S.: Domain Theory in Logical Form. In: Proceedings of the Second Annual IEEE Symposium on Logic in Computer Science (LICS 1987), pp. 47–53. IEEE Computer Society Press (1987)
3. Abramsky, S.: Domain theory in logical form. Annals of Pure and Applied Logic **51**(1), 1–77 (1991). https://doi.org/10.1016/0168-0072(91)90065-T
4. Birkmann, F., Urbat, H., Milius, S.: Monoidal Extended Stone Duality. In: Kobayashi, N., Worrell, J. (eds.) Foundations of Software Science and Computation Structures. LNCS, vol. 14574, pp. 144–165. Springer, Heidelberg (2024). https://doi.org/10.1007/978-3-031-57228-9_8
5. Blackburn, P., de Rijke, M., Venema, Y.: Modal Logic. Cambridge University Press (2001)
6. Chagrov, A., Zakharyaschev, M.: Modal Logic. Oxford University Press (1996)
7. Cignoli, R., Lafalce, S., Petrovich, A.: Remarks on Priestley duality for distributive lattices. Order **8**(3), 299–315 (1991). https://doi.org/10.1007/BF00383451
8. Davey, B.A., Priestley, H.A.: Introduction to Lattices and Order. Cambridge University Press (2002)
9. Dijkstra, E.W.: A Discipline of Programming. Prentice-Hall (1976)
10. Esakia, L.: Heyting Algebras: Duality Theory. Springer International Publishing (2019)
11. Halmos, P.R.: Algebraic logic, I. Monadic boolean algebras. Compositio Mathematica **12**, 217–249 (1956). https://www.numdam.org/item/CM_1954-1956__12__217_0/
12. Hansoul, G.: A duality for Boolean algebras with operators. Algebra Universalis **17**(1), 34–49 (1983). https://doi.org/10.1007/BF01194512
13. Hennessy, M.C.B., Plotkin, G.D.: Full abstraction for a simple parallel programming language. In: Bečvář, J. (ed.) Mathematical Foundations of Computer Science 1979. LNCS, vol. 74, pp. 108–120. Springer, Heidelberg (1979). https://doi.org/10.1007/3-540-09526-8_8
14. Heunen, C., Vicary, J.: Categories for Quantum Theory: An Introduction. Oxford University Press (2019)
15. Hoare, C.A.R.: An axiomatic basis for computer programming. Communications of the ACM **12**(10), 576–580 (1969). https://doi.org/10.1145/363235.363259
16. Hofmann, D., Nora, P.: Dualities for modal algebras from the point of view of triples. Algebra universalis **73**(3), 297–320 (2015). https://doi.org/10.1007/s00012-015-0324-5
17. Hofmann, D., Nora, P.: Duality theory for enriched Priestley spaces. Journal of Pure and Applied Algebra **227**(3), 107231 (2023). https://doi.org/10.1016/j.jpaa.2022.107231
18. Jacobs, B.: Introduction to Coalgebra: Towards Mathematics of States and Observation. Cambridge University Press (2016)

19. Johnstone, P.T.: Stone Spaces. Cambridge University Press (1982)
20. Jonsson, B., Tarski, A.: Boolean Algebras with Operators. Part I. American Journal of Mathematics **73**(4), 891 (1951). https://doi.org/10.2307/2372123
21. Kavvos, G.A.: Two-Dimensional Kripke Semantics I: Presheaves. In: Rehof, J. (ed.) 9th International Conference on Formal Structures for Computation and Deduction (FSCD 2024). Leibniz International Proceedings in Informatics (LIPIcs), 14:1–14:23. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl, Germany (2024). https://doi.org/10.4230/LIPIcs.FSCD.2024.14
22. Kavvos, G.A.: Two-dimensional Kripke Semantics II: Stability and Completeness. Electronic Notes in Theoretical Informatics and Computer Science **Volume 4 - Proceedings of MFPS XL** (2024). https://doi.org/10.46298/entics.proceedings.mfps40
23. Kishida, K.: Categories and Modalities. In: Categories for the Working Philosopher. Ed. by E. Landry. Oxford University Press (2018). https://doi.org/10.1093/oso/9780198748991.003.0009. (Visited on 2022-03-29)
24. Kripke, S.: Semantical Considerations on Modal Logic. Acta Philosophica Fennica **16**, 83–94 (1963). https://doi.org/10.1002/malq.19630090502
25. Kripke, S.A.: Semantical Analysis of Intuitionistic Logic I. In: ed. by J.N. Crossley and M.A.E. Dummett, pp. 92–130. Elsevier (1965). https://doi.org/10.1016/S0049-237X(08)71685-9. (Visited on 01/28/2023)
26. Kripke, S.A.: Semantical Analysis of Modal Logic I. Normal Modal Propositional Calculi. Zeitschrift für Mathematische Logik und Grundlagen der Mathematik **9**(5-6), 67–96 (1963). https://doi.org/10.1002/malq.19630090502
27. Kurz, A., Moshier, A., Jung, A.: Stone Duality for Relations. In: Samson Abramsky on Logic and Structure in Computer Science and Beyond. Ed. by A. Palmigiano and M. Sadrzadeh, pp. 159–215. Springer International Publishing, Cham (2023). https://doi.org/10.1007/978-3-031-24117-8_5
28. Lawvere, F.W.: Equality in Hyperdoctrines and Comprehension Schema as an Adjoint Functor. In: Proceedings of the AMS Symposium on Pure Mathematics, pp. 1–14 (1970). https://ncatlab.org/nlab/files/LawvereComprehension.pdf
29. Melliès, P.-A., Zeilberger, N.: A bifibrational reconstruction of Lawvere's presheaf hyperdoctrine. In: Proceedings of the 31st Annual ACM/IEEE Symposium on Logic in Computer Science, pp. 555–564. Association for Computing Machinery, New York NY USA (2016). https://doi.org/10.1145/2933575.2934525
30. Nielsen, M., Plotkin, G., Winskel, G.: Petri nets, event structures and domains, Part I. Theoretical Computer Science **13**(1), 85–108 (1981). https://doi.org/10.1016/0304-3975(81)90112-2
31. Plotkin, G.D.: A powerdomain construction. SIAM Journal on Computing **5**(3), 452–487 (1976)
32. Sambin, G., Vaccaro, V.: Topology and duality in modal logic. Annals of Pure and Applied Logic **37**(3), 249–296 (1988). https://doi.org/10.1016/0168-0072(88)90021-8
33. Sangiorgi, D.: On the origins of bisimulation and coinduction. ACM Transactions on Programming Languages and Systems **31**(4), 1–41 (2009). https://doi.org/10.1145/1516507.1516510
34. Scott, D.S.: Relating Theories of the Lambda Calculus. In: To H. B. Curry: Essays on Combinatory Logic, Lambda Calculus, and Formalism. Ed. by J.P. Seldin and J.R. Hindley. Academic Press, London (1980)
35. Shulman, M.A.: Framed bicategories and monoidal fibrations. Theory and Applications of Categories **20**(18), 650–738 (2008)

36. Thijs, A.M.: Simulation and Fixpoint Semantics. PhD thesis, University of Groningen (1996). https://hdl.handle.net/11370/13d08025-29ff-4193-a7f2-ea5bcd20f15d.
37. Thomason, S.K.: Categories of Frames for Modal Logic. The Journal of Symbolic Logic **40**(3), 439–442 (1975). https://doi.org/10.2307/2272167
38. Wittgenstein, L.: Tractatus Logico-Philosophicus. Kegan Paul, Trench, Trubner & Co., Ltd., London (1922)